



Generative Artificial Intelligence and Large Language Models in Rehabilitation and Disability Services: A Scoping Review

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ABSTRACT

The rapid emergence of generative artificial intelligence (GenAI) tools and large language models (LLMs) since late 2022 has introduced new possibilities for rehabilitation and disability services. However, no review has systematically mapped the scope and nature of evidence on GenAI applications specifically in these contexts. This scoping review aimed to map the evidence on GenAI and LLM applications in rehabilitation and disability services, examining the tools applied and their purposes (SQ1), populations and service contexts represented (SQ2), reported outcomes, benefits, and concerns (SQ3), and evidence gaps with implications for future research, practice, and policy (SQ4). Sources published from late 2022 to May 2026 in English, addressing the application of GenAI or LLMs within rehabilitation, disability, or special education service contexts, involving persons with disabilities, rehabilitation professionals, special educators, or related stakeholders. Scopus, Web of Science, and Lens were searched in May 2026. Screening reliability was assessed computationally (Cohen's $\kappa = 0.938$). Data were extracted using Elicit across 10 variables and reviewed by the principal investigator. Synthesis followed a narrative and descriptive approach organised by the four sub-questions. Ninety-six studies published between 2023 and 2026 were included. ChatGPT was referenced in 85.4% of studies. The dominant purposes were assistive and adaptive technology (40.6%), professional training (21.9%), and content generation (21.9%). Higher education rehabilitation programmes (51.0%) and school-level special education (28.1%) were the most common service contexts. Learning disabilities (17.7%), autism spectrum disorder (14.6%), and visual impairment (11.5%) were the most specified disability types, though 56.2% of studies did not specify a disability diagnosis. Reported benefits included measurable improvements in learning outcomes under structured, educator-guided conditions, significant time savings for special educators and rehabilitation professionals, and enhanced autonomy for students with disabilities. Concerns included clinical reasoning accuracy limitations, AI-generated accessibility failures, readability exceeding health literacy thresholds, and equity barriers in low-resource settings. Critical gaps included the near-total absence of longitudinal outcome data, very limited direct patient-facing clinical evidence, sparse representation of complex disability profiles, and no governance or institutional policy research. The evidence supports cautious, educator-guided integration of GenAI in rehabilitation and special education training, with structured pedagogical design and human oversight as critical conditions for benefit. The evidence does not yet support conclusions about patient outcomes, long-term competency development, or population-level equity effects. Future research should prioritise longitudinal designs, direct patient-facing clinical studies, under-represented disability populations, and disability-inclusive AI governance frameworks.

KEYWORDS

Generative artificial intelligence, Large language models, ChatGPT, Rehabilitation, Disability, Special education, Scoping review, Assistive technology, Equity

INTRODUCTION

Disability affects approximately 15% of the global population, with prevalence rates higher among women, older adults, and individuals in low- and middle-income countries (Mitra and Sambamoorthi, 2014; Olusanya *et al.*, 2020). Access to rehabilitation services—essential for optimising functional

capacity, promoting independence, and enabling social participation—remains constrained by financial, structural, and systemic barriers worldwide (Sousa *et al.*, 2017; Clemente *et al.*, 2022; Kayola *et al.*, 2023). These barriers include high costs of transportation and treatment, insufficient numbers

of trained rehabilitation specialists, geographic remoteness of services, and persistent social stigma surrounding disability (Kayola *et al.*, 2023; Chimienti *et al.*, 2026; Jasim *et al.*, 2026). The gap between the need for and availability of rehabilitation services is particularly acute in low-resource settings, where the rehabilitation workforce and infrastructure are most limited (Sousa *et al.*, 2017; Kayola *et al.*, 2023).

The World Health Organization's Rehabilitation 2030 initiative has formally recognised this unmet need, calling for the integration of rehabilitation into universal health coverage and health system strengthening worldwide (World Health Organization, 2017). Within this agenda, disability-inclusive digital health has emerged as a priority mechanism for extending service reach, but digital tools carry a dual potential: they can improve access, or, when designed without disability inclusion, introduce new barriers and exclusion (Henni *et al.*, 2022; van Kessel *et al.*, 2022). Generative artificial intelligence (GenAI) applications in this space span a spectrum from assistive technologies that support communication, learning, and independence directly for persons with disabilities to clinical decision support tools that inform professional judgement; these serve distinct purposes and carry distinct risk profiles, a distinction maintained throughout this review.

The emergence of AI has introduced new possibilities for addressing some of these longstanding service gaps. Prior to 2022, AI applications in rehabilitation were concentrated primarily in robotics-assisted motor rehabilitation, wearable sensor-based monitoring, and rule-based decision support systems (Vélez-Guerrero, Callejas-Cuervo and Mazzoleni, 2021; Sumner *et al.*, 2023; Subbarayalu *et al.*, 2024). Systematic reviews of this earlier evidence base documented promising applications in stroke care, physical therapy, and orthopaedic rehabilitation, but consistently identified limited high-quality clinical evidence, under-representation of certain disability groups, and the absence of long-term outcome data (Sumner *et al.*, 2023; Estes-Schmalzl and Lefebvre, 2025; Sadiq *et al.*, 2026). In special education, AI was applied primarily through intelligent tutoring systems, adaptive learning platforms, and robotic aids, with reviews highlighting potential benefits for personalised learning and communication support alongside persistent barriers in teacher training, digital access, and ethical governance (Hopcan *et al.*, 2023; Hussein, Hussein and Al-Hendawi, 2025; Li, Yan and Zeng, 2025).

The public release of ChatGPT in November 2022 marked a qualitative shift in the AI landscape. For the first time, a generative AI (GenAI) tool capable of producing fluent natural language text, engaging in extended dialogue, and responding to complex professional queries became freely accessible to clinicians, educators, students, and persons with disabilities without specialised technical skills (Boscardin *et al.*, 2024; Chen, Liu and Zhu, 2024). The rapid proliferation of competing large language models (LLMs)—including Gemini, Claude, Llama, DeepSeek, Bard, and Copilot—has since created an ecosystem of GenAI tools with potential applications across clinical care, professional training, content generation, assistive technology, and service administration in rehabilitation and disability contexts (Rabbani *et al.*, 2025; Zhang, Tan and Lin, 2025; Zhang *et al.*, 2025).

Early evidence from healthcare and education more broadly suggests that GenAI can enhance clinical decision-making, reduce administrative burden, improve learning engagement, and support personalised instruction, while simultaneously raising concerns about hallucinations, misinformation, privacy risks, academic integrity, and an emerging digital divide that may disproportionately affect vulnerable populations (Shrivastava, 2024; Rabbani *et al.*, 2025; Qian, 2025; Nakagomi, Abe and Tabuchi, 2026).

These terms are related but not interchangeable: LLMs are text-based foundation models trained on large text corpora; multimodal GenAI systems (e.g. GPT-4o, Gemini) extend this to process and generate images, audio, or video alongside text; and 'foundation model' refers more broadly to any large pre-trained model adaptable to multiple downstream tasks, of which LLMs are one type. This review uses 'GenAI' as the umbrella term and specifies LLM, multimodal, or foundation model characteristics where relevant to the included evidence.

Despite this rapid growth, the intersection of GenAI with rehabilitation and disability services has received limited dedicated synthesis. Existing reviews have focused on AI broadly in rehabilitation (Sumner *et al.*, 2023; Sadiq *et al.*, 2026) or in special education (Hopcan *et al.*, 2023; Hussein, Hussein and Al-Hendawi, 2025), but no review has systematically mapped the scope and nature of evidence on GenAI and LLMs specifically in rehabilitation and disability service contexts. Adjacent reviews have addressed related but distinct questions: Zhu *et al.* (2025) conducted a Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews/Population, Concept, and Context scoping review of GenAI specifically in stroke rehabilitation, and Xu *et al.* (2026) examined AI interaction design for neurodivergent users. The present review differs from these by addressing the full breadth of disability types, rehabilitation disciplines, service contexts, and stakeholder groups rather than a specific condition or user group. This gap is significant because the distinctive capabilities and risks of GenAI—particularly its capacity for natural language interaction, content generation, and accessibility support—differ fundamentally from those of earlier AI technologies and require dedicated evidence synthesis. A scoping review is particularly appropriate for this task given the rapidly evolving nature of the evidence base, the heterogeneity of study designs and populations, and the aim to map the breadth of available evidence rather than to assess the effectiveness of specific interventions (Munn *et al.*, 2018, 2022; Pollock *et al.*, 2024). Moreover, questions of equity and access in digital health technologies are especially pressing for disability populations, who face compounded barriers including inaccessible design, cost constraints, limited digital literacy, and structural exclusion from technology development processes (Henni *et al.*, 2022; van Kessel *et al.*, 2022).

Objectives

The aim of this scoping review was to systematically map the scope and nature of the evidence on the application of GenAI tools and LLMs in rehabilitation and disability

services. Using the Population, Concept, and Context (PCC) framework (Aromataris *et al.*, 2024), the review addressed the following primary question: What is the scope and nature of the evidence on the application of GenAI tools and LLMs in rehabilitation and disability services?

Four sub-questions guided the review. SQ1: What types of GenAI tools and LLMs have been applied or studied across rehabilitation and disability service settings, and for what purposes? SQ2: What populations (disability types, age groups, stakeholder roles) and service contexts are represented in the evidence? SQ3: What outcomes, benefits, or concerns related to accessibility, service quality, and equity have been reported? SQ4: What gaps exist in the evidence base, and what are the implications for future research, practice, and policy?

METHOD

Protocol and registration

This scoping review was conducted in accordance with the JBI Manual for Evidence Synthesis, Chapter 10 (Aromataris *et al.*, 2024), and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR; Tricco *et al.*, 2018). The completed PRISMA-ScR checklist is provided in Supplementary File S3. The scoping review design was selected as the most appropriate evidence synthesis approach given the broad, emerging nature of the topic and the aim to map the scope and nature of available evidence rather than to assess the effectiveness of specific interventions (Munn *et al.*, 2018, 2022). Scoping reviews are particularly suited to rapidly evolving fields where evidence is heterogeneous and where a comprehensive conceptual map of the literature is needed to inform future research priorities and practice development (Khalil *et al.*, 2021; Pollock *et al.*, 2022). A formal review protocol was developed and agreed upon by the research team prior to commencement of searching. The protocol was not prospectively registered in a public repository. Prospective registration is not a mandatory requirement of JBI or PRISMA-ScR guidance for scoping reviews (Tricco *et al.*, 2018), and it was not undertaken in this instance; nonetheless, the complete search strategy, eligibility criteria, and data charting framework are reported in full in Supplementary File S1 and openly archived via OSF to ensure methodological transparency and replicability. All three supplementary files are openly available via the Open Science Framework (OSF) at https://osf.io/s3tgz/overview?view_only=14a1cbcb558e459689ce2314b9625491.

Eligibility criteria

Eligibility criteria were developed using the PCC framework recommended by JBI for scoping reviews (Aromataris *et al.*, 2024). The population element encompassed persons with disabilities (any type, any age group); rehabilitation professionals including physiotherapists, occupational therapists,

speech-language pathologists, and psychologists; special educators; service administrators and policymakers working in disability or rehabilitation contexts; and family caregivers of persons with disabilities. At least one of these groups was required to be directly involved in or addressed by each eligible source. The concept element encompassed the application, use, development, implementation, evaluation, or governance of GenAI tools or LLMs in rehabilitation or disability contexts, including clinical, educational, community-based, tele-rehabilitation, assistive technology, professional training, caregiver support, and ethical or governance applications. Named GenAI tools—including ChatGPT, Gemini, Claude, Llama, DeepSeek, Bard, and Copilot—are illustrative examples of the broader concept; the review was not restricted to these named tools. The context element encompassed any rehabilitation or disability service setting, including healthcare, educational, community-based, tele-rehabilitation, and policy contexts, with no geographic restriction.

Sources were eligible if published from January 2022 to May 2026, coinciding with the emergence of widely accessible post-ChatGPT GenAI tools (i.e. LLMs became publicly available only in late 2022). Searches were restricted to English-language sources; the inability to include non-English publications, including Korean-language studies identified during searching, constitutes a limitation of the review. All peer-reviewed source types were eligible, including empirical studies (quantitative, qualitative, and mixed methods), reviews, case studies, conference papers (full text available), book chapters (full text available), and conceptual or theoretical papers. Preprints, editorials, letters, errata, datasets, meeting abstracts, correction notices, and sources for which full text was not accessible were excluded. Sources addressing rehabilitation exclusively in non-disability medical contexts (e.g. cardiac, pulmonary, or substance use rehabilitation) without a disability population were also excluded.

Information sources

Search strategy

The search strategy was developed iteratively through a series of pilot searches and comprised three conceptual blocks connected by the Boolean operator AND. Block 1 encompassed GenAI and LLM terms (e.g. ‘generative AI’, ‘large language model*’, ‘ChatGPT’, ‘Gemini’, ‘Claude’, ‘Llama’, ‘DeepSeek’, ‘Bard’, ‘Copilot’, ‘foundation model*’). Block 2 encompassed rehabilitation and disability service terms (e.g. ‘rehabilitation’, ‘disabilit*’, ‘special education’, ‘speech therapy’, ‘occupational therapy’, ‘physiotherapy’, ‘assistive communication’, ‘augmentative and alternative communication’). Block 3 encompassed outcomes, workforce, and implementation terms (e.g. ‘acceptance’, ‘barrier*’, ‘facilitator*’, ‘training’, ‘service delivery’, ‘equity’, ‘clinical practice’). All three blocks were applied exclusively to the Title field across all databases to ensure that both GenAI and disability/rehabilitation were central rather than incidental foci of retrieved sources. This decision

was made following a preliminary broader search applying Title-Abstract-Key word fields, which retrieved 790 records; a preliminary relevance scan confirmed substantial noise from studies mentioning GenAI or disability only tangentially, supporting the adoption of a more precise Title-field strategy that privileged topical centrality over comprehensiveness. Studies for which GenAI or disability are discussed primarily in the abstract or keywords rather than the title may therefore have been missed, which is acknowledged as a scope limitation of the search strategy. The full search string for each database is reported in Supplementary File S1.

Selection of sources of evidence

Records exported from all three databases in CSV format were merged and deduplicated computationally using Python (version 3.12; Python Software Foundation, Wilmington, DE, USA) with the pandas library. Deduplication was conducted in two stages: DOI-based matching followed by title-based matching for records without a DOI. Following deduplication, document-type exclusions were applied (preprints, editorials, datasets, letters, errata, and meeting abstracts), yielding 138 records for title and abstract screening.

Title and abstract screening was conducted in two parallel stages. First, a rule-based screening algorithm was applied computationally (Claude Sonnet 4; Anthropic, 2025) against the confirmed inclusion and exclusion criteria, with uncertain cases reviewed manually by the principal investigator. Second, results were verified using Elicit (2025), an AI-assisted systematic review platform, with the confirmed eligibility criteria entered as structured screening instructions. Screening reliability was assessed by comparing decisions across the principal investigator and Elicit, yielding an AI-assisted screening concordance of $\kappa = 0.938$ (almost perfect agreement; Landis and Koch, 1977), with an overall percentage agreement of 98.4%. Full Cohen's κ values and pairwise comparisons for all three decision-making agents are reported in Supplementary File S1, Section 9. It should be noted that this statistic reflects concordance between a human reviewer and an AI platform, not between two independent human reviewers (the conventional use of Cohen's κ in evidence synthesis). A three-way percentage agreement of 82.0% was computed across the three decision-making agents: Claude Sonnet 4 (Stage 1 computational screening), Elicit (Stage 2 AI-assisted verification), and the principal investigator (final arbitration). Apparent disagreements were attributable entirely to full-text accessibility failures and duplicate record identification rather than divergent eligibility judgements. All discrepancies were resolved through final arbitration by the principal investigator. This approach to AI-assisted screening is consistent with emerging guidance on the use of AI tools in evidence synthesis (Khalil *et al.*, 2021; Pollock *et al.*, 2023).

In addition to the three decision-making agents described above, two faculty colleagues with expertise in systematic evidence synthesis and AI applications in education conducted face-validity reviews of all major methodological stages—including the search strategy, eligibility criteria, screening decisions, and data extraction framework—prior

to finalisation. These face-validity reviews provided an additional layer of human expert verification across all stages of the review. They were not included in the kappa calculation as they did not constitute independent blind screening, but they substantively strengthened the methodological rigour and quality assurance of the review process.

Records passing title and abstract screening were uploaded to Elicit for full-text screening. Full texts were retrieved manually by the principal investigator where Elicit was unable to access the full text automatically. Records for which full text could not be retrieved in English were excluded ($n = 18$). Conference papers were evaluated against the same eligibility criteria as journal articles and were eligible only when the full conference paper text was retrievable; conference abstracts without accessible full texts were excluded at this stage. Three additional records were excluded at the data verification stage upon identification as duplicate or supplementary file records. A final set of 96 studies was included in the review.

Several methodological considerations specific to the AI-assisted screening workflow warrant explicit discussion. Regarding prompt standardisation: the eligibility criteria were entered into Elicit as a structured six-criterion screening instruction, with each criterion defined as a pass/fail rule, and the combined rubric was piloted on ten records before full screening. The Claude Sonnet 4 algorithm applied the same criteria as a rule-based filter. Regarding model version stability: both Claude Sonnet 4 (Anthropic, 2025) and Elicit (2025) are commercial platforms subject to version updates; exact replication cannot be guaranteed, as model behaviours may change. Regarding automation bias: the sequential design—in which the PI reviewed uncertain cases flagged by Stage 1 and then compared them against Elicit at Stage 2—carries a risk that systematic errors shared by both AI systems could propagate undetected; this was mitigated by PI personal review of all full-text eligibility decisions. Regarding ethical implications: accountability for all screening decisions rests with the PI as the human arbitrator. AI-generated screening outputs were not archived beyond the Elicit platform interface; researchers seeking to replicate the screening process are advised to contact the corresponding author.

Data charting process

Data charting was conducted using Elicit (2025) in accordance with JBI guidance on the extraction, analysis, and presentation of results in scoping reviews (Pollock *et al.*, 2023). A structured charting form comprising 10 extraction fields was developed collaboratively by the research team prior to data extraction and is reported in full in Supplementary File S1. The charting form was piloted on a random sample of five included studies prior to full extraction, and field definitions were refined to ensure consistent interpretation across source types. Elicit's LLM extracted data from each full-text source in accordance with the structured field instructions. All extracted data were reviewed by the principal investigator for accuracy, completeness, and consistency, and corrections were made where extraction was incomplete or imprecise.

Data were not extracted in duplicate by a second independent reviewer, which constitutes a limitation of the charting process acknowledged by Khalil *et al.* (2021) as a common challenge in independently conducted scoping reviews. The single-reviewer pattern extended to all stages of the review: the principal investigator alone arbitrated all AI screening disagreements, made all full-text retrieval decisions, and reviewed all Elicit extractions. While the AI-assisted workflow was designed to provide decision checks, no stage of this review involved independent validation by a second human reviewer—a limitation for replication and verification. No contact with original study authors was made to confirm or supplement extracted data.

Data items

Ten variables were extracted from each included source in accordance with recommendations for scoping review data presentation (Pollock *et al.*, 2023). (1) Study design and methodology: study type, specific method where applicable, sample size, and country or region. (2) GenAI tool(s) examined: tool name(s) as stated, version where specified, or ‘General LLM/GenAI—not specified’ where no specific tool was named. (3) Purpose or function of GenAI: categorised as assessment or diagnosis support, intervention or therapeutic tool, assistive or adaptive technology, content generation or instructional material creation, professional training or education, administrative or documentation support, research or analytical tool, ethical or policy analysis, or accessibility enhancement. (4) Target population: disability type(s), age group, and stakeholder role. (5) Service or study context: setting category and country or region. (6) Key findings and outcomes: primary outcome, reported benefits, and reported concerns or negative findings. (7) Equity and accessibility considerations: coded as ‘reported’ or ‘not reported’ against specified criteria including digital access barriers, algorithmic bias, affordability, and data privacy. (8) Stakeholder perspectives: stakeholder group(s) whose perspectives were reported. (9) Limitations reported by authors. (10) Research gaps and future directions identified by authors. Full definitions and extraction instructions for each variable are provided in Supplementary File S1.

Critical appraisal of individual sources of evidence

Consistent with JBI scoping review methodology (Aromataris *et al.*, 2024) and the conceptual distinction between scoping reviews and systematic reviews of effectiveness (Munn *et al.*, 2018), formal critical appraisal of individual sources of evidence was not conducted. Scoping reviews aim to map the breadth of available evidence rather than to assess its quality for the purpose of clinical recommendations; accordingly, all eligible sources were included regardless of methodological quality. Methodological limitations identified by study authors themselves were charted as a data item and are synthesised in the discussion of evidence gaps. The heterogeneity of

study designs, the predominance of small and single-site studies, and the near-complete absence of longitudinal data are noted as characteristics of this emerging evidence base rather than grounds for exclusion.

Synthesis of results

Throughout the synthesis, conceptual and commentary papers were narratively flagged as such and were not accorded the same evidentiary weight as empirical findings; they were drawn upon to characterise perspectives, proposed frameworks, and identified concerns rather than as evidence of measured outcomes.

Data synthesis followed a narrative and descriptive approach consistent with JBI scoping review guidance (Peters *et al.*, 2020; Aromataris *et al.*, 2024) and recommendations for the extraction, analysis, and presentation of results in scoping reviews (Pollock *et al.*, 2023). Charted data were first summarised descriptively using frequency counts and percentages for categorical variables as presented in Table 1. Findings were then synthesised narratively, organised according to the four sub-questions (SQ1-SQ4) guiding the review. Within each sub-question, studies were grouped thematically according to shared findings, populations, or contexts, and convergent and divergent findings were identified. An illustrative example of the scoping review approach applied to a clinical population may be found in Alduais *et al.* (2022). No quantitative meta-analysis was attempted given the heterogeneity of study designs, outcome measures, and populations across included sources. Quantitative findings from individual studies are reported as presented by original authors without pooling or re-analysis.

RESULTS

Selection of sources of evidence

The three-phase search strategy identified 306 records across Scopus ($n = 94$), Web of Science ($n = 71$), and Lens.org ($n = 141$). Following removal of 150 duplicate records and 18 records excluded at the document type stage, 138 records proceeded to title and abstract screening. Twenty-one records were excluded at this stage, leaving 117 records for full-text retrieval. Of these, 18 could not be retrieved due to inaccessible or non-English full text. Three records were subsequently excluded at the data verification stage as duplicate or supplementary file records, yielding a final included set of 96 studies. The complete selection process is presented in Figure 1.

Characteristics of sources of evidence

The 96 included studies were published between 2023 and 2026, with the majority appearing in 2025 ($n = 49$, 51.0%)

Table 1: Characteristics of included studies ($N = 96$).

Characteristic	Category	<i>n</i> (%)
Publication year	2023	3 (3.1)
	2024	15 (15.6)
	2025	49 (51.0)
	2026	29 (30.2)
Source type	Journal article	74 (77.1)
	Conference paper	18 (18.8)
	Book chapter	2 (2.1)
	Review article	2 (2.1)
Study design	Survey/questionnaire	29 (30.2)
	Qualitative	16 (16.7)
	Quasi-experimental	11 (11.5)
	Conceptual/commentary	12 (12.5)
	Mixed methods	9 (9.4)
	Other/not specified	9 (9.4)
	Secondary review	3 (3.1)
	RCT	3 (3.1)
	Descriptive/cross-sectional	2 (2.1)
	Experimental	1 (1.0)
	Case study	1 (1.0)
GenAI tool(s) examined ^a	ChatGPT/GPT (any version)	82 (85.4)
	Gemini	19 (19.8)
	Claude	11 (11.5)
	Bard	10 (10.4)
	Copilot	9 (9.4)
	General LLM/not specified	24 (25.0)
	DeepSeek	5 (5.2)
	Llama	3 (3.1)
Purpose or function of GenAI ^a	Assistive/adaptive technology	39 (40.6)
	Professional training	21 (21.9)
	Content generation/instructional materials	21 (21.9)
	Intervention/therapeutic tool	12 (12.5)
	Accessibility enhancement	6 (6.2)
	Assessment/diagnosis support	3 (3.1)
	Administrative/documentation	3 (3.1)
	Research/analytical tool	3 (3.1)
Service or study context	Higher education (rehabilitation/special education programs)	49 (51.0)
	Special education—school level	27 (28.1)
	Clinical/healthcare setting	9 (9.4)
	Conceptual/no specific setting	4 (4.2)
	Community-based services	3 (3.1)
	Tele-rehabilitation/digital platform	3 (3.1)
	Other/mixed	1 (1.0)
Disability type or educational context ^a	Special education/disability context not further specified ^b	54 (56.2)
	Learning disabilities/dyslexia	17 (17.7)
	Autism spectrum disorder	14 (14.6)
	Visual impairment	11 (11.5)
	Intellectual disabilities	8 (8.3)
	Hearing impairment/deafness	8 (8.3)
	Physical/motor disability	7 (7.3)
	ADHD	5 (5.2)
	Cerebral palsy	5 (5.2)
	Stroke	4 (4.2)

Table 1: Continued.

Characteristic	Category	<i>n</i> (%)
Stakeholder group ^a	Neurological conditions	3 (3.1)
	Aphasia	1 (1.0)
	Dementia/cognitive decline	1 (1.0)
	Rehabilitation/special education students	46 (47.9)
	Persons with disabilities	41 (42.7)
	Special educators/teachers	35 (36.5)
	Rehabilitation professionals	30 (31.2)
	Policymakers/administrators	9 (9.4)
	Caregivers/parents	7 (7.3)
	No participants (conceptual)	3 (3.1)
Geographic region ^c	Not reported in study	48 (50.0)
	USA	10 (10.4)
	Saudi Arabia	6 (6.2)
	China	5 (5.2)
	Japan	4 (4.2)
	Türkiye	3 (3.1)
	Greece	3 (3.1)
	Egypt	3 (3.1)
	United Kingdom	3 (3.1)
	Sweden	2 (2.1)
	Other (nine countries, <i>n</i> = 1 each)	9 (9.4)
	Yes	73 (76.0)
	Not reported	23 (24.0)

Note: A complete study-by-study charting table is provided in Supplementary File S2. Abbreviations: ADHD, attention deficit hyperactivity disorder; GenAI, generative artificial intelligence; LLM, large language model; RCT, randomised controlled trial.

^aPercentages exceed 100% because individual studies may report multiple tools, populations, functions, or stakeholder groups.

^bStudies coded as “special education/disability context not further specified” addressed disability or special education as their primary context but did not identify a specific disability type or diagnosis (e.g. studies focused on special education teachers, disability services broadly, or mixed disability populations without disaggregated reporting). All 96 studies are directly relevant to disability or rehabilitation by virtue of their inclusion in the review.

^cGeographic region was determined from the institutional affiliation(s) of the corresponding author(s) as reported in the source record. The nine countries represented by a single study each are Australia, Brazil, Kenya, Nigeria, Palestine, Philippines, South Korea, Spain, and Tanzania.

^dEquity and accessibility considerations were coded as “reported” when the study explicitly addressed at least one of the following: digital access barriers, cultural or linguistic equity, affordability of AI tools, algorithmic bias toward disability, or data privacy concerns for vulnerable populations.

and 2026 (*n* = 29, 30.2%), reflecting the rapid growth of this field following the public release of ChatGPT in late 2022. Journal articles constituted the largest source type (*n* = 74, 77.1%), followed by conference papers (*n* = 18, 18.8%), book chapters (*n* = 2, 2.1%), and review articles (*n* = 2, 2.1%). A comprehensive summary of study characteristics is presented in Table 1, illustrative examples are provided in Table 2, and the complete data charting table for all 96 included studies is provided in Supplementary File S2.

Survey and questionnaire designs were the most prevalent study methodology (*n* = 29, 30.2%), followed by qualitative approaches (*n* = 16, 16.7%), quasi-experimental designs (*n* = 11, 11.5%), and conceptual or commentary papers (*n* = 12, 12.5%). Mixed-methods studies accounted for 9.4% (*n* = 9), and three randomised controlled trials (RCTs) were identified. Sample sizes ranged from 4 to over 800,000, though most fell below 500 participants and

many intervention studies were notably small (*n* < 50). The geographic distribution spanned 20 countries, with the most represented regions being the United States (*n* = 10), Saudi Arabia (*n* = 6), China (*n* = 5), Japan (*n* = 4), and Türkiye (*n* = 3); 48 studies (50.0%) did not report the country of origin.

The methodological landscape of the included evidence warrants contextualisation for readers interpreting the review’s findings. Empirical studies predominated (*n* = 76, 79.2%), encompassing quantitative surveys (*n* = 29), qualitative studies (*n* = 16), quasi-experimental designs (*n* = 11), mixed-methods studies (*n* = 9), descriptive/cross-sectional studies (*n* = 2), experimental studies (*n* = 1), case studies (*n* = 1), and three RCTs. Conceptual and commentary papers (*n* = 12, 12.5%), secondary reviews (*n* = 3, 3.1%), and book chapters (*n* = 2, 2.1%) constituted the remainder. Across the empirical studies, sample sizes were predominantly modest: most intervention studies enrolled fewer than 50 participants,

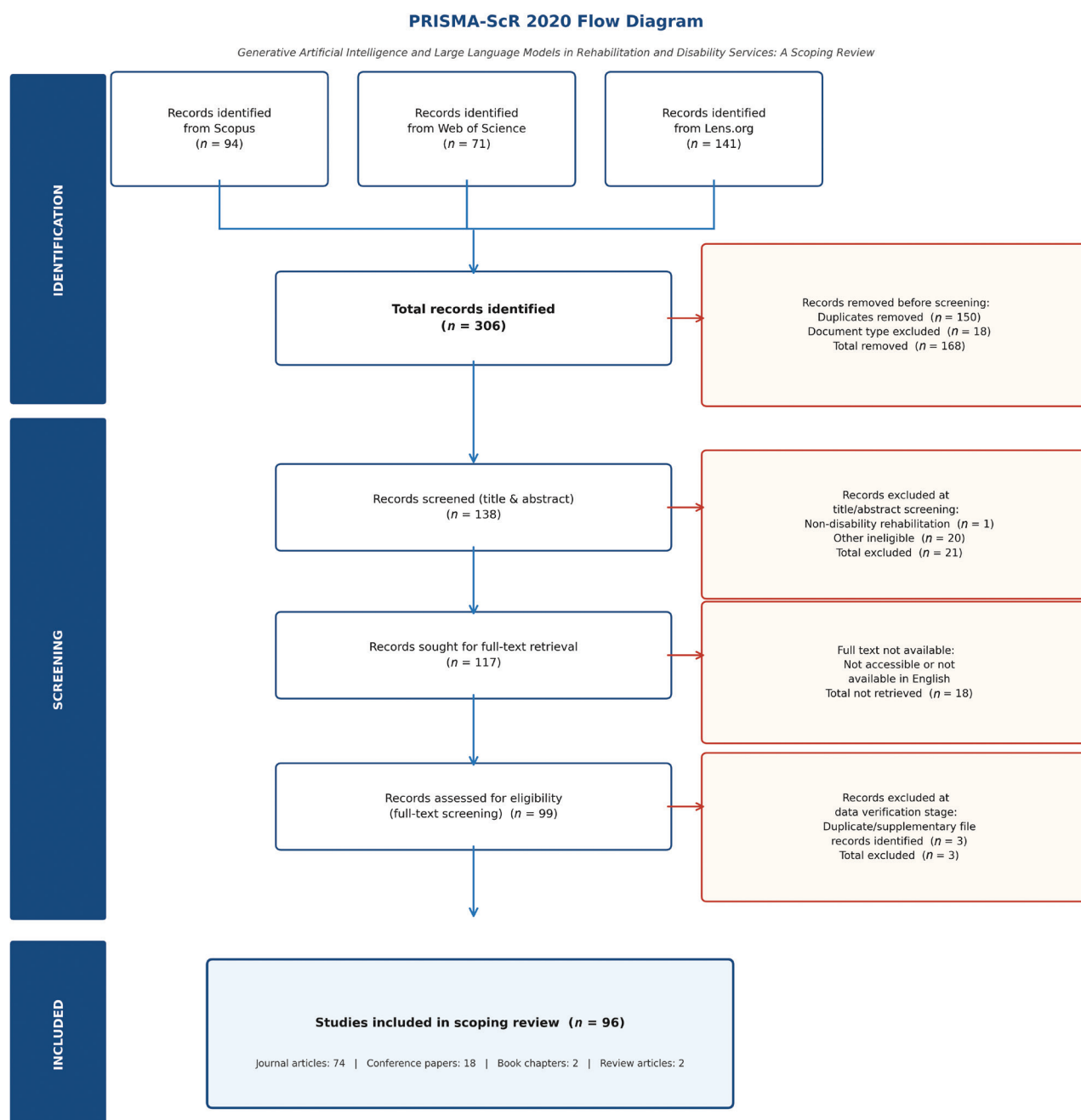


Figure 1: PRISMA-ScR 2020 flow diagram. Note. Screening reliability: Title/abstract screening was conducted computationally (Claude Sonnet 4; Anthropic, 2025) and verified using Elicit (Elicit, 2025). Agreement between the principal investigator and Elicit: AI-assisted screening concordance: $\kappa = 0.938$ (almost perfect agreement; Landis and Koch, 1977). Three-way agreement (Claude Sonnet 4, Elicit, PI): 82.0%. All discrepancies resolved by principal investigator arbitration. Chart adapted from Page *et al.* (2021) and reported in accordance with Tricco *et al.* (2018).

and only a small minority exceeded 300. The three RCTs are the most methodologically rigorous designs in the corpus but involve small, single-site samples and examine time-bounded tasks. The predominance of surveys, qualitative studies, and conceptual papers indicates that the evidence base remains largely descriptive and attitudinal in nature, with limited causal inference capacity. These characteristics should be borne in mind when interpreting the synthesis: the findings reflect the scope and topics of this emerging field rather than constituting a quality-weighted evaluation of intervention effectiveness.

SQ1: GenAI tools applied and their purposes

Tool landscape

ChatGPT—spanning versions 3.5, 4, 4o, and 4.5—was referenced in 82 of 96 included studies (85.4%), functioning across virtually every application category in the review (Alenezi, Mohamed and Shaaban, 2023; Lyerly, 2023a,b; Ibrahim and Ajlouni, 2024; Mansour and Wong, 2024; Neo, Ser and Tay, 2024; Seiradakis, 2024, 2025; Shahzad *et al.*, 2024; Atcheson *et al.*, 2025; Cao, 2025; Clancy *et al.*, 2025;

Table 2: Illustrative examples of included studies by source type.

Study	GenAI tool(s)	Disability/population	Setting and stakeholders	Key finding
Okuno <i>et al.</i> (2025)	ChatGPT, GPT-4	Disability not specified (physiotherapy students)	Higher education Physiotherapy students	An LLM-generated story-centred curriculum produced significantly greater improvements in proficiency test scores than historical controls, with positive learner experience responses exceeding 90% across all domains.
Khlaif <i>et al.</i> (2025)	ChatGPT	Visual impairment	Special education (higher) Students with disabilities	ChatGPT enhanced personalised learning and real-time content access for visually impaired university students, promoting educational equity and reducing reliance on human intermediaries.
Elkot <i>et al.</i> (2025)	ChatGPT	Mild intellectual disabilities	Special education (school) Students with intellectual disabilities	Structured AI-guided conversation significantly improved English communication skills across vocabulary, grammar, pronunciation, fluency, and comprehension, with guided conditions outperforming unstructured use.
Atcheson <i>et al.</i> (2025)	ChatGPT, DALL-E, Google Bard	Disability not specified (multiple disability types)	Special education (higher) Students with disabilities	University students with disabilities valued GenAI for personalising learning, promoting self-care, and assisting self-advocacy, describing it as more consistently available than formal disability services.
Yu <i>et al.</i> (2026)	GPT-4	Aphasia	Clinical/healthcare Patients, speech therapists, family caregivers	An LLM-integrated embodied digital therapist for aphasia rehabilitation reduced therapist time per patient by 69-78%, with patient acceptance improving by 42.6% across sessions and greatest gains among low digital literacy users.
Elavsky and Bearfield (2025)	ChatGPT (4o), Claude (Sonnet 4)	Visual impairment (blind/low-vision)	Conceptual/no specific setting Persons with visual disabilities	Integrating generative models into accessibility workflows introduced “verification disability”—blind and low-vision users could not verify AI-generated descriptions and were compelled to rely on potentially inaccurate outputs without accountability mechanisms.
de Azevedo <i>et al.</i> (2025)	ChatGPT	Disability not specified (physiotherapy students)	Higher education Physiotherapy students, educators	AI-generated clinical case studies enhanced physiotherapy education through integrated AI, offering richer scenarios and reducing educator preparation time while supporting diverse learning styles.
Damianidou (2025)	ChatGPT	Autism spectrum disorder	Special education (school) Special educators	ChatGPT supported educator understanding of student needs and well-being in ASD contexts, offering tailored instructional strategies and improving teacher confidence in inclusive practice.
Voultsiou <i>et al.</i> (2025)	ChatGPT, GPT-4, LLaMA, Bard, PaLM 2	ASD, hearing impairment, cerebral palsy, learning disabilities	Special education (school) Students with disabilities, educators	Systematic review found strong potential for LLM-integrated social robots in special education across communication, social skills, and academic domains, while identifying persistent barriers in teacher training and infrastructure.
Severin and Gagnon (2025)	ChatGPT (general)	Disability not specified (physical therapy educators/students)	Higher education Physical therapy educators, administrators, students	Only 35.4% of physical therapy educators and students reported using generative AI, suggesting the field remains in an exploratory phase; structured exposure rather than awareness alone was identified as the key adoption driver.

Note: Studies were selected purposively to illustrate the breadth of the evidence base across source types, GenAI tools, disability populations, service settings, and geographic regions. All studies are from the 96 included in this review. Source types are indicated in brackets within the ‘Study’ column: blue, journal article; yellow, conference paper; green, book chapter; pink, review article. The complete data charting table for all 96 included studies are provided in Supplementary File S2. Abbreviations: AAC, augmentative and alternative communication; ASD, autism spectrum disorder; GenAI, generative artificial intelligence; IEP, individualised education programme; LLM, large language model.

Damianidou, 2025; Dekerlegand *et al.*, 2025; Durgungoz and Durgungoz, 2025; Elkot *et al.*, 2025; El-Sobkey *et al.*, 2025, 2026; Hao, Yao and Siu, 2025; Khlaif *et al.*, 2025; Lee and Joo, 2025; Lin, Han and Qiu, 2025; Lindbäck *et al.*, 2025a,b; Okuno, Kawamitsu and Yamaguchi, 2025; Palmer and Oswal, 2025; Pierrès, Darvishy and Christen, 2025; Qiang *et al.*, 2025; Safran and Yildirim, 2025; Sakowicz and Hamidi, 2025; Severin and Gagnon, 2025; Tom, Wu and Shea, 2025; van Garderen *et al.*, 2025; Zhao, Cox and Chen, 2025; Alqarni, 2026a,b; Balikci, Terzioglu and Rakap, 2026; Değerli, Özata Değerli and Kars, 2026; Fletcher, 2026; Güler *et al.*, 2026; Habila, Mwendaga and Victor, 2026; Lin, Cui and Zhu, 2026; Meny, 2026; Naatz and Ruppert, 2026; Rappa *et al.*, 2026; Santos *et al.*, 2026; Seung and Basham, 2026; Sigirtmac *et al.*, 2026; Yu *et al.*, 2026). Gemini appeared in a significant secondary cluster ($n = 19$, 19.8%), most notably in comparative evaluations (Othman *et al.*, 2024; Elshaer, Alnajdi and Salem, 2025; Kilinç and Altıntaş, 2025; Michou *et al.*, 2026; Severin and Gagnon, 2025; Saita, Mine and Amano, 2026). Claude featured in 11 studies (11.5%), Bard in 10 (10.4%), Copilot in 9 (9.4%), DeepSeek in 5 (5.2%), and Llama in 3 (3.1%). A meaningful proportion of studies ($n = 24$, 25.0%) examined GenAI in general terms without specifying a particular tool (Alshaigy and Grande, 2024; Cao, 2025; Lindbäck *et al.*, 2025b; Lee and Joo, 2025; Lopez-Gazpio, 2025; Wairimu, 2025; Fletcher, 2026; Kim, Lee and Song, 2026; Santos *et al.*, 2026). Several studies examined locally developed or domain-adapted models, including ApAI (Kilinç and Altıntaş, 2025), MedGo and ERNIE Bot (Qiang *et al.*, 2025), and a custom ChatGPT-based question-generating bot for occupational therapy licensure preparation (Kohiyama *et al.*, 2025).

Purpose categories

Seven primary purposes were identified. Professional training and education was one of the most common purposes ($n = 21$, 21.9%), encompassing physiotherapy (Al-Fishawy, 2025; Cao, 2025; El-Sobkey *et al.*, 2025, 2026; Kohiyama *et al.*, 2025; Lindbäck *et al.*, 2025a,b; Okuno, Kawamitsu and Yamaguchi, 2025; Severin and Gagnon, 2025; Güler *et al.*, 2026; Michou *et al.*, 2026), occupational therapy (Mansour and Wong, 2024; Villagrán *et al.*, 2024; Sudo, Noborimoto and Takahashi, 2025; Tom, Wu and Shea, 2025; Meny, 2026; Saita, Mine and Amano, 2026; Santos *et al.*, 2026; Sigirtmac *et al.*, 2026), and special education preservice training (Ibrahim and Ajlouni, 2024; Shahzad *et al.*, 2024; Durgungoz and Durgungoz, 2025). Content generation and instructional material creation were the second most common purposes ($n = 21$, 21.9%), applied in special education for IEP objectives and lesson plans (Smith, Smith and Price, 2024; Lin, Han and Qiu, 2025; Lombardi *et al.*, 2025; Sakowicz and Hamidi, 2025; van Garderen *et al.*, 2025; Lin, Cui and Zhu, 2026; Naatz and Ruppert, 2026) and in rehabilitation professional training for clinical case studies and patient education materials (de Azevedo *et al.*, 2025; Hao, Yao and Siu, 2025; Qiang *et al.*, 2025; Sari and Çelik, 2026; Sari, Çelik and Mirza, 2026; Zorlular and Zorlular, 2026). Assistive and adaptive technology applications ($n = 39$,

40.6%) encompassed sign language translation (Othman *et al.*, 2024), AAC enhancement (Zhang, Tan and Lin, 2025; Zhang *et al.*, 2025), and accessibility support for students with visual, hearing, learning, and cognitive disabilities (Alenezi, Mohamed and Shaaban, 2023; Lyerly, 2023a,b; Alshaigy and Grande, 2024; Husin *et al.*, 2024; Nacheva, 2024; Damianidou, 2025; Elshaer, AlNajdi and Salem, 2025; Farhah *et al.*, 2025; Francisco, Hervás and García-Mata, 2025; Jenks, Onufer and Guberman, 2025; Khlaif *et al.*, 2025; Lopez-Gazpio, 2025; Voultsiou and Moussiades, 2025; Wairimu, 2025; Zhao, Cox and Chen, 2025; Aidarus, Lamers and Khribi, 2026; Alqarni, 2026a,b; Habila, Mwendaga and Victor, 2026; Rappa *et al.*, 2026; Seung and Basham, 2026). Intervention and therapeutic applications ($n = 12$, 12.5%) included AI-supported communication training for students with intellectual disabilities (Elkot *et al.*, 2025), ChatGPT for motivation enhancement (Alkenani *et al.*, 2025), and LLM-driven digital therapists for aphasia rehabilitation (Yu *et al.*, 2026). Accessibility enhancement ($n = 6$, 6.3%; Nacheva, 2024; Elavsky and Bearfield, 2025; Palmer and Oswal, 2025), administrative and documentation support ($n = 3$, 3.1%; Clancy *et al.*, 2025; Habila, Mwendaga and Victor, 2026), and assessment and clinical performance evaluation ($n = 3$, 3.1%; Drossinou Korea and Alexopoulos, 2024; Safran and Yildirim, 2025; Tang *et al.*, 2025; Saita, Mine and Amano, 2026) were the smallest purpose categories. Studies using GenAI as a research or analytical tool ($n = 3$, 3.1%) included ChatGPT-assisted categorisation of open-ended survey responses in a Parkinson's disease rehabilitation study (Conte *et al.*, 2024), disability ethics analysis in clinical rehabilitation education (Urbina, Vu and Nguyen, 2025), and rehabilitation exercise quality assessment using prompt engineering (Tang *et al.*, 2025).

SQ2: Populations and service contexts

Disability types and age groups

The evidence base spans a wide but uneven range of disability types. Learning disabilities, dyslexia, and attention deficit hyperactivity disorder constituted the largest group ($n = 17$; Alenezi, Mohamed and Shaaban, 2023; Drossinou Korea and Alexopoulos, 2024; Alkenani *et al.*, 2025; Farhah *et al.*, 2025; Kuerban, Oyelere and Sanusi, 2025; Lopez-Gazpio, 2025; van Garderen *et al.*, 2025; Voultsiou and Moussiades, 2025; Wang *et al.*, 2025; Zhao, Cox and Chen, 2025; Alqarni, 2026a,b; Habila, Mwendaga and Victor, 2026; Naatz and Ruppert, 2026; Seung and Basham, 2026). Autism spectrum disorder (ASD) featured prominently ($n = 14$; Seiradakis, 2024; Damianidou, 2025; Wang *et al.*, 2025; Balikci, Terzioglu and Rakap, 2026; Rappa *et al.*, 2026). Visual impairment received dedicated attention ($n = 11$; Lyerly, 2023a,b; Elavsky and Bearfield, 2025; Elshaer, AlNajdi and Salem, 2025; Khlaif *et al.*, 2025; Kilinç and Altıntaş, 2025; Palmer and Oswal, 2025; Anselimus, 2026). Intellectual disabilities were directly studied in several intervention studies ($n = 8$; Phalaguna, Kaewsaeng and Worabuttara, 2024; Elkot *et al.*, 2025; Lee and Joo, 2025; Rappa *et al.*, 2026). Hearing impairment was examined across sign language translation,

conversational AI, and student experience studies ($n = 8$; Nacheva, 2024; Othman *et al.*, 2024; Pierrès, Darvishy and Christen, 2025; Anselimus, 2026). Physical and motor disabilities, stroke, cerebral palsy, aphasia, and neurological conditions were represented primarily in clinical rehabilitation contexts (Neo, Ser and Tay, 2024; Hao, Yao and Siu, 2025; Qiang *et al.*, 2025; Değerli, Özata Değerli and Kars, 2026; Sari and Çelik, 2026; Sari, Çelik and Mirza, 2026; Xiong *et al.*, 2026; Yu *et al.*, 2026; Zorlular and Zorlular, 2026). Fifty-four studies (56.2%) addressed disability or special education broadly without specifying a disability type or diagnosis.

Stakeholder groups

Students in rehabilitation or special education programmes were the most frequently represented stakeholder group ($n = 46$, 47.9%), appearing across physiotherapy (Shahzad *et al.*, 2024; Al-Fishawy, 2025; El-Sobkey *et al.*, 2025, 2026; Lindbäck *et al.*, 2025a,b; Okuno, Kawamitsu and Yamaguchi, 2025; Güler *et al.*, 2026; Kim, Lee and Song, 2026; Michou *et al.*, 2026), occupational therapy (Mansour and Wong, 2024; Dekerlegand *et al.*, 2025; Kohiyama *et al.*, 2025; Tom, Wu and Shea, 2025; Meny, 2026; Santos *et al.*, 2026; Sigirtmac *et al.*, 2026), and special education (Ibrahim and Ajlouni, 2024; Durgungoz and Durgungoz, 2025; Severin and Gagnon, 2025) programmes. Special education teachers formed the second largest group ($n = 35$, 36.5%; Seiradakis, 2024; Lin, Han and Qiu, 2025; Sakowicz and Hamidi, 2025; van Garderen *et al.*, 2025; Alqarni, 2026a,b; Balikci, Terzioglu and Rakap, 2026; Lin, Cui and Zhu, 2026; Naatz and Rupp, 2026; Habila, Mwendaga and Victor, 2026). Rehabilitation professionals appeared in 30 studies (31.3%; Arbel, Gimmon and Shmueli, 2025; Cao, 2025; Clancy *et al.*, 2025; El-Sobkey *et al.*, 2025; Hao, Yao and Siu, 2025; Safran and Yildirim, 2025; Severin and Gagnon, 2025; Değerli, Özata Değerli and Kars, 2026; Kim, Lee and Song, 2026). Persons with disabilities as primary participants appeared in 41 studies (42.7%; Neo, Ser and Tay, 2024; Atcheson *et al.*, 2025; Damianidou, 2025; Elavsky and Bearfield, 2025; Elshaer, AlNajdi and Salem, 2025; Khlaif *et al.*, 2025; Palmer and Oswal, 2025; Pierrès, Darvishy and Christen, 2025; Zhao, Cox and Chen, 2025; Anselimus, 2026). Family caregivers appeared in seven studies (7.3%; Neo, Ser and Tay, 2024; Phalaguna, Kaewsaeng and Worabuttara, 2024; Qiang *et al.*, 2025; Wang *et al.*, 2025; Xiong *et al.*, 2026; Yu *et al.*, 2026; Zorlular and Zorlular, 2026), and policymakers or administrators in nine studies (9.4%; Alshaigy and Grande, 2024; Clancy *et al.*, 2025; Kiliç and Altıntaş, 2025; Severin and Gagnon, 2025; Lin, Han and Qiu, 2026).

Service contexts

University-based rehabilitation programmes accounted for the largest single service context cluster ($n = 49$, 51.0%; Mansour and Wong, 2024; Shahzad *et al.*, 2024; Al-Fishawy, 2025; Arbel, Gimmon and Shmueli, 2025; Cao, 2025; Dekerlegand *et al.*, 2025; El-Sobkey *et al.*, 2025, 2026;

Kohiyama *et al.*, 2025; Lindbäck *et al.*, 2025a,b; Okuno, Kawamitsu and Yamaguchi, 2025; Severin and Gagnon, 2025; Tom, Wu and Shea, 2025; Güler *et al.*, 2026; Kim, Lee and Song, 2026; Meny, 2026; Michou *et al.*, 2026; Santos *et al.*, 2026; Sigirtmac *et al.*, 2026). Special education at school level was the second most common context ($n = 27$, 28.1%; Alenezi, Mohamed and Shaaban, 2023; Drossinou Korea and Alexopoulos, 2024; Othman *et al.*, 2024; Phalaguna, Kaewsaeng and Worabuttara, 2024; Seiradakis, 2024, 2025; Alkenani *et al.*, 2025; Damianidou, 2025; Farhah *et al.*, 2025; Kuerban, Oyelere and Sanusi, 2025; Lee and Joo, 2025; Lin, Han and Qiu, 2025; Sakowicz and Hamidi, 2025; van Garderen *et al.*, 2025; Voultsiou and Moussiades, 2025; Alqarni, 2026a,b; Balikci, Terzioglu and Rakap, 2026; Habila, Mwendaga and Victor, 2026; Lin, Cui and Zhu, 2026; Naatz and Rupp, 2026; Rappa *et al.*, 2026). Special education in higher education (Lyerly, 2023a,b; Ibrahim and Ajlouni, 2024; Nacheva, 2024; Atcheson *et al.*, 2025; Durgungoz and Durgungoz, 2025; Elshaer, AlNajdi and Salem, 2025; Khlaif *et al.*, 2025; Pierrès, Darvishy and Christen, 2025; Wairimu, 2025; Zhao, Cox and Chen, 2025; Alqarni, 2026a; Anselimus, 2026; Fletcher, 2026; Liu *et al.*, 2026), clinical rehabilitation settings ($n = 9$; Neo, Ser and Tay, 2024; Hao, Yao and Siu, 2025; Qiang *et al.*, 2025; Değerli, Özata Değerli and Kars, 2026; Sari, Çelik and Mirza, 2026; Xiong *et al.*, 2026; Yu *et al.*, 2026; Zorlular and Zorlular, 2026), tele-rehabilitation (Kohiyama *et al.*, 2025; Zhang, Tan and Lin, 2025; Zhang *et al.*, 2025; Sari and Çelik, 2026), and community-based disability services (Palmer and Oswal, 2025; Alofi, Al-Shammrani and Hammad, 2026) each accounted for smaller proportions.

SQ3: Outcomes, benefits, and concerns

Benefits and positive outcomes

Several intervention studies reported measurable improvements in learning outcomes. In the methodologically strongest study in the corpus, an LLM-generated story-centred curriculum for physiotherapy students produced significantly greater improvements in proficiency test scores compared to historical controls, with positive questionnaire responses exceeding 90% across all learner experience domains (Okuno, Kawamitsu and Yamaguchi, 2025). Guided GenAI conversation significantly improved English communication skills across vocabulary, grammar, pronunciation, fluency, and comprehension in students with mild intellectual disabilities, with structured guidance producing superior outcomes over unstructured use (Elkot *et al.*, 2025). ChatGPT similarly supported the development of functional writing skills—specifically email composition—in students with intellectual disabilities through a self-monitoring instructional approach, with benefits including enhanced error correction and writing accuracy (Li and Liu, 2024). A ChatGPT-supported motivation intervention for students with learning disabilities reported a large effect size ($\eta^2 = 0.405$; Alkenani *et al.*, 2025), and writing skill improvements were observed in a quasi-experimental study using an experience-based AI writing programme for students with intellectual disabilities (Lee and Joo, 2025). LLMs demonstrated

competitive performance on structured professional knowledge assessments, with all four tested achieving passing scores of 82.4–91.0% on the Japanese National Examination for Occupational Therapists (Saita, Mine and Amano, 2026) and producing higher-quality written responses than final-year physiotherapy students across four clinical domains (Michou *et al.*, 2026).

Time savings represented one of the most consistently reported practical benefits. An RCT found that ChatGPT-assisted task analysis significantly reduced special education teacher preparation time (Balıkcı, Terzioğlu and Rakap, 2026). An LLM-integrated digital therapist for aphasia rehabilitation reduced therapist time per patient by 69–78%, with patient acceptance improving by 42.6% across sessions (Yu *et al.*, 2026). Special education teachers across multiple contexts reported consistent time savings in lesson planning and IEP goal writing (Phalaguna, Kaewsang and Worabuttara, 2024; Smith, Smith and Price, 2024; Lin, Han and Qiu, 2025; Lombardi *et al.*, 2025; Sakowicz and Hamidi, 2025; van Garderen *et al.*, 2025; Alqarni, 2026a,b; Lin, Cui and Zhu, 2026; Naatz and Ruppert, 2026).

Students with disabilities in higher education consistently described GenAI as enhancing their autonomy and reducing reliance on intermediaries (Lyerly, 2023a,b; Atcheson *et al.*, 2025; Elshaer, AlNajdi and Salem, 2025; Khlaif *et al.*, 2025; Pierrès, Darvishy and Christen, 2025; Zhao, Cox and Chen, 2025; Anselimus, 2026; Fletcher, 2026; Zhao, Chen and Cox, 2026). Early conceptual literature identified broad potential across learning, communication, and independence goals (Alshaigy and Grande, 2024; Lopez-Gazpio, 2025; Voultsiou and Moussiades, 2025; Wairimu, 2025). Early childhood special education experts identified ChatGPT as holding significant assistive potential while flagging ethical and accuracy risks (Seiradakis, 2024). Faculty members reported positive perceptions of ChatGPT's role in supporting students with learning disabilities in inclusive higher education (Alqarni, 2026a). OT students perceived ChatGPT positively as an academic tool for content generation and review (Meny, 2026). Fieldwork readiness in OT students was reported to improve through structured GenAI engagement (Mansour and Wong, 2024). AI-generated gamified quizzes were perceived positively as an engaging learning tool (Durgungoz and Durgungoz, 2025). ChatGPT improved the well-being and understanding of the needs of students with ASD when used by teachers in structured instructional contexts (Damianidou, 2025). Systematic reviews confirmed GenAI's most common benefits as communication support, behaviour management, and academic skill development across diverse disabilities (Voultsiou and Moussiades, 2025; Wang *et al.*, 2025), with particular promise in resource-constrained settings when appropriately adapted (Wairimu, 2025). A bottom-up policy analysis of GenAI affordances for students with disabilities in UK higher education identified three core affordances—explainability, expressibility, and plannability—that positively affected learning processes for students with neurodiversity, dyslexia, and social/communication impairments and highlighted the need for student-informed institutional GenAI policies (Zhao, Chen and Cox, 2026).

Concerns and negative findings

Accuracy limitations and AI hallucinations were the most uniformly reported concern. ChatGPT demonstrated sub-optimal clinical reasoning, failing to tailor examination strategies to individual patient profiles, omitting red flag screening, and missing pharmacological factors—raising explicit patient safety concerns (Hao, Yao and Siu, 2025). ChatGPT recommendations in occupational therapy scenarios did not fully align with clinical reasoning and suggested inappropriate assessment tools (Değerli, Özata Değerli and Kars, 2026). In stroke patient education, ChatGPT and Bard hallucinated responses including fabricated information about support resources (Neo, Ser and Tay, 2024). Readability of AI-generated patient education content consistently exceeded recommended health literacy thresholds (Sari and Çelik, 2026; Sari, Çelik and Mirza, 2026; Zorlular and Zorlular, 2026). LLMs showed a marked decline from 82.4–91.0% accuracy on text-based questions to only 50.0–68.2% on visually based items in occupational therapy licensing examinations (Saita, Mine and Amano, 2026). A critical literature review identified hallucination, bias, and lack of clinical contextualisation as systemic concerns in LLM applications across special education and training (Drossinou Korea and Alexopoulos, 2024).

Concerns about over-reliance and erosion of professional skill development emerged consistently. A post-test-only RCT found no significant improvement in cognitive, affective, or psychomotor outcomes without structured educator guidance (Sigirtmac *et al.*, 2026). Conceptual analysis identified the risk of over-reliance eroding skill development for students with learning disabilities (Seung and Basham, 2026). Physiotherapy students described surface learning risks and uncertain use boundaries (Lindbäck *et al.*, 2025b). A UK higher education perspective argued that positioning GenAI as a technological solution constitutes 'technological solutionism', potentially relieving institutional obligations to address structural support deficits (Fletcher, 2026). Ethical implications of GenAI as assistive technology for students with learning disabilities were examined across contexts, with teachers reporting concerns about academic integrity, bias, and appropriate boundaries (Seiradakis, 2025; Habila, Mwendaga and Victor, 2026).

GenAI tools simultaneously introduced new accessibility barriers. AI-generated websites failed basic screen-reader accessibility requirements (Palmer and Oswal, 2025), and a study on data visualisation identified 'verification disability' whereby blind and low-vision users could not verify AI-generated descriptions without accountability mechanisms (Elavsky and Bearfield, 2025). In sub-Saharan Africa, structural constraints including high subscription costs and inadequate infrastructure significantly limited GenAI's accessibility benefits (Anselimus, 2026). Early reviews noted that the lack of disability-specific design and training data remained a structural barrier to equitable deployment (Alshaigy and Grande, 2024; Lopez-Gazpio, 2025). Adoption remained partial and context-dependent across professional groups; in the largest clinical rehabilitation professional survey ($n = 264$), 61.1% did not use GenAI during the study period (Clancy *et al.*, 2025), and only 35.4%

of PT educators and students in the United States reported using GenAI (Severin and Gagnon, 2025). Across all contexts, experiential familiarity predicted acceptance more strongly than professional role or demographic characteristics (Alenezi, Mohamed and Shaaban, 2023; Dekerlegand *et al.*, 2025; Güler *et al.*, 2026; Kim, Lee and Song, 2026).

SQ4: Evidence gaps and implications for future research, practice, and policy

Four categories of evidence gaps emerged consistently. First, the near-complete absence of longitudinal data is the most salient methodological gap. No study tracked participants beyond the immediate post-intervention period (Dekerlegand *et al.*, 2025; Farhah *et al.*, 2025; Kohiyama *et al.*, 2025; Okuno, Kawamitsu and Yamaguchi, 2025; Balikci, Terzioglu and Rakap, 2026; Güler *et al.*, 2026; Naatz and Ruppar, 2026; Rappa *et al.*, 2026; Yu *et al.*, 2026). Most intervention studies were small, single-site, and relied on self-report measures (Ibrahim and Ajlouni, 2024; Mansour and Wong, 2024; Alkenani *et al.*, 2025; Atcheson *et al.*, 2025; El-Sobkey *et al.*, 2025; Lee and Joo, 2025; Sigirtmac *et al.*, 2026). Future priorities include RCTs and longitudinal cohort studies (Hao, Yao and Siu, 2025; Safran and Yildirim, 2025; Yu *et al.*, 2026), controlled studies on GenAI impact on IEP quality over academic years (Sakowicz and Hamidi, 2025; Lin, Cui and Zhu, 2026; Naatz and Ruppar, 2026), and mixed-methods studies tracking adoption trajectories (Clancy *et al.*, 2025; Dekerlegand *et al.*, 2025; Kim, Lee and Song, 2026).

Second, the gap between GenAI use in professional education, where evidence is most abundant, and its use in direct patient care, where evidence is sparse, is the most strategically important. Only a handful of studies engaged directly with patients as users (Neo, Ser and Tay, 2024; Qiang *et al.*, 2025; Xiong *et al.*, 2026; Yu *et al.*, 2026), with universally small samples and no clinical outcome data beyond satisfaction or engagement. Third, several disability populations remain under-represented, including traumatic brain injury, spinal cord injury, progressive neurological conditions, complex communication needs, and multiple concurrent disabilities (Wairimu, 2025; Wang *et al.*, 2025; Zhang, Tan and Lin, 2025; Zhang *et al.*, 2025), as well as younger children and older adults with disabilities (Pierrès, Darvishy and Christen, 2025; Anselimus, 2026). No included study addressed psychiatric or mental health disabilities, deafblindness, or degenerative conditions other than dementia, indicating that the evidence base, while broad, remains concentrated in learning, developmental, sensory, and physical disability categories rather than representing the full spectrum of disability recognised in rehabilitation practice.

Fourth, equity and governance dimensions are raised by many studies but rigorously examined by few. Evidence on GenAI use in low-resource and low- and middle-income country rehabilitation settings is limited to a single qualitative study (Wairimu, 2025; Anselimus, 2026). No study examined institutional governance responses to GenAI integration in rehabilitation or disability services. The call for involving persons with disabilities as co-creators in AI

tool development and policy appeared repeatedly but had not translated into participatory research designs within this corpus (Atcheson *et al.*, 2025; Elavsky and Bearfield, 2025; Palmer and Oswal, 2025; Fletcher, 2026). Early conceptual work consistently identified the need for disability-inclusive AI governance frameworks and ethical oversight (Alshaigy and Grande, 2024; Drossinou Korea and Alexopoulos, 2024; Cao, 2025; Souval *et al.*, 2025; Lopez-Gazpio, 2025; Urbina, Vu and Nguyen, 2025; Habila, Mwendaga and Victor, 2026).

DISCUSSION

It should be emphasised at the outset that this evidence base is overwhelmingly educational and training-focused rather than clinical: the large majority of included studies concern physiotherapy, occupational therapy, and special education training contexts, with direct patient-facing rehabilitation evidence remaining sparse (see SQ4). Conclusions regarding clinical rehabilitation practice should accordingly be distinguished from those regarding professional and student education, which the evidence supports far more directly.

Summary of evidence

This scoping review mapped the scope and nature of evidence on the application of GenAI tools and LLMs in rehabilitation and disability services across 96 studies published between 2023 and 2026. The findings reveal a rapidly expanding but methodologically immature evidence base dominated by ChatGPT, concentrated in higher education rehabilitation training and school-level special education, and characterised by a consistent tension between documented benefits and systematic concerns.

These findings also warrant engagement with disability rights principles beyond accessibility narrowly defined. Autonomy, supported decision-making, and communication access are central tenets of disability rights frameworks, and the reviewed evidence suggests that GenAI can advance or undermine each depending on how it is deployed. Direct, user-controlled use by persons with disabilities for writing, reading, and self-organisation supports autonomy and self-determination (Atcheson *et al.*, 2025; Khlaif *et al.*, 2025; Anselimus, 2026), whereas automated deployment of GenAI on behalf of disabled users without their participation risks the opposite: decisions and content generated about them rather than with them, echoing longstanding disability rights critiques of paternalistic technology design. Digital exclusion remains a live risk throughout, since the populations most likely to benefit from GenAI-supported communication access are also those most exposed to cost, connectivity, and digital literacy barriers (Henni *et al.*, 2022; Anselimus, 2026).

This review also differs in kind, not only in scope, from earlier AI-in-rehabilitation syntheses. Prior reviews mapped AI applications centred on robotics, wearable sensors, and decision-support algorithms in stroke and physical rehabilitation

(Vélez-Guerrero, Callejas-Cuervo and Mazzoleni, 2021; Sumner *et al.*, 2023; Sadiq *et al.*, 2026). GenAI is not simply another entrant in this same category: its language-based capabilities fundamentally change how users interact with technology, how professionals document and communicate, and how accessibility support is delivered—through natural language dialogue, on-demand content generation, and text-based communication access rather than sensor-driven measurement or physical assistance. This distinction explains why the benefits and risks identified in this review (e.g. accessibility gains alongside accuracy and bias concerns) differ qualitatively from those reported for earlier, non-GenAI applications in rehabilitation.

Regulatory and governance frameworks emerging alongside this evidence base are directly relevant to its future application. The European Union's Artificial Intelligence Act [Regulation (EU) 2024/1689] classifies AI systems by risk level and imposes transparency and human-oversight obligations that would apply to GenAI tools used in educational or clinical rehabilitation contexts. The World Health Organization's guidance on large multimodal models in health (World Health Organization, 2024) similarly calls for inclusive design, post-deployment auditing, and disaggregated impact assessment by user characteristics, including disability—recommendations directly applicable to the populations addressed in this review. Neither framework was reflected in the included literature, underscoring a governance gap consistent with SQ4.

The near-complete dominance of ChatGPT (85.4% of included studies) reflects its first-mover advantage and free accessibility following the November 2022 public release (Boscardin *et al.*, 2024; Chen, Liu and Zhu, 2024). This concentration carries both practical and methodological implications: findings from ChatGPT-based studies may not generalise to other LLMs, and the rapid versioning cycle of commercial models means that studies evaluating GPT-3.5 may have limited relevance to current GPT-4o or GPT-5 capabilities. The meaningful proportion of studies (25.0%) that examined GenAI in general terms without specifying a tool version further complicates cross-study comparison and replicability. This echoes broader concerns about the reproducibility of GenAI research in healthcare (Rabbani *et al.*, 2025) and education (Shrivastava, 2024).

The evidence presents a coherent pattern regarding educational applications. The strongest positive findings—the LLM-generated story-centred curriculum in physiotherapy (Okuno, Kawamitsu and Yamaguchi, 2025), the guided conversational AI for students with intellectual disabilities (Elkot *et al.*, 2025), and the ChatGPT-supported motivation intervention for students with learning disabilities (Alkenani *et al.*, 2025)—all involved deliberate pedagogical design with defined task structures and human oversight. By contrast, the RCT testing chatbot-assisted case-based learning without structured pedagogical framing found no significant difference in outcomes (Sigirtmac *et al.*, 2026). This pattern is consistent with the theoretical framework of cognitive offloading: GenAI delivers benefits when it reduces task-irrelevant cognitive load, freeing capacity for higher-order processing, but may undermine learning when it eliminates the cognitive engagement that

drives skill development (Seung and Basham, 2026). The implication is that the modality and structure of GenAI integration, not simply its presence, determines educational outcomes.

The clinical accuracy evidence converges on a coherent picture of capability alongside systematic gaps. LLMs performed well on knowledge retrieval, text synthesis, and structured question-answering—tasks that map to foundational didactic knowledge (Safran and Yildirim, 2025; Michou *et al.*, 2026; Saita, Mine and Amano, 2026)—but demonstrated consistent limitations in nuanced clinical reasoning that requires integrating patient-specific context, screening for red flags, and adapting to individual presentations (Hao, Yao and Siu, 2025; Değerli, Özata Değerli and Kars, 2026). This is not a binary failure but a domain-specific competency boundary: LLMs appear reliable as knowledge-retrieval and first-pass educational tools but should not be treated as substitutes for practitioner-level clinical reasoning in rehabilitation contexts. The patient safety implications of this boundary are most clearly articulated in the musculoskeletal physical therapy evaluation, where ChatGPT's failure to screen for red flags in spinal cases represents a concrete, foreseeable risk if outputs are used without expert oversight (Hao, Yao and Siu, 2025). These findings align with broader evidence that LLMs in healthcare demonstrate strong textual competence but limited clinical contextualisation (Rabbani *et al.*, 2025; Zhang, Tan and Lin, 2025; Zhang *et al.*, 2025).

The divergence between accessibility benefits reported by persons with disabilities in higher education and the accessibility failures identified in technical evaluations of AI-generated products reflects two distinct populations and use cases. Students with disabilities who actively choose to use ChatGPT for writing, reading support, and self-organisation report substantial perceived benefits (Atcheson *et al.*, 2025; Khlaif *et al.*, 2025; Pierrès, Darvishy and Christen, 2025; Anselimus, 2026). These benefits are real for users with the digital literacy, language skills, and device access to engage with current LLM interfaces. However, when GenAI is deployed to automatically produce accessible products for others—such as accessible websites (Palmer and Oswal, 2025) or data visualisation descriptions for blind users (Elavsky and Bearfield, 2025)—the absence of disability-focused training data and accountability mechanisms introduces new risks of exclusion. Both findings are valid within their respective contexts and point towards the same practical implication: direct use of LLMs by informed, digitally literate users with disabilities can be beneficial, while automated deployment of LLMs to solve accessibility problems on behalf of disabled users without their participation introduces new risks of bias and exclusion. This finding resonates with broader evidence on the digital health paradox, wherein technologies designed to improve health equity can inadvertently widen disparities for persons with disabilities when deployed without inclusive design (Henni *et al.*, 2022; van Kessel *et al.*, 2022).

The marked variation in GenAI adoption and acceptance across professional groups and national contexts is best explained by experiential familiarity rather than individual-level attitudes alone. Across all contexts, actual

users consistently reported higher perceived usefulness, lower concern, and higher future intention than non-users (Alenezi, Mohamed and Shaaban, 2023; Dekerlegand *et al.*, 2025; Güler *et al.*, 2026; Kim, Lee and Song, 2026). This pattern has direct implications for professional development policy: structured, scaffolded exposure to GenAI tools—rather than awareness campaigns alone—is likely the most effective strategy for promoting informed and critical adoption among rehabilitation professionals and special educators. The finding that 61.1% of US rehabilitation professionals did not use GenAI (Clancy *et al.*, 2025) and that only 35.4% of physical therapy educators reported using it (Severin and Gagnon, 2025) suggests that the field remains in an early exploratory phase, with substantial room for growth alongside a clear need for institutional guidance.

Directly comparing findings across studies reveals meaningful variation that the descriptive synthesis cannot fully capture. Across the three RCTs in the corpus, results diverge in instructive ways: the task analysis RCT (Balıkcı, Terzioğlu and Rakap, 2026) found significant time savings for special education teachers—a structured, practitioner-facing, content-generation task with clearly bounded scope—while the case-based learning RCT in occupational therapy (Sigirtmac *et al.*, 2026) found no significant improvement in clinical reasoning outcomes in a more open-ended, higher-order learning task. This divergence supports the hypothesis that the specificity and structure of the task, rather than the presence of GenAI alone, determines whether measurable benefits emerge. Across disability groups, the contrast is also notable: studies of learning disabilities, ASD, and intellectual disabilities—where structured, repetitive, and scaffolded practice is theoretically well motivated—reported more consistent positive outcomes than studies of complex clinical populations (stroke, aphasia, musculoskeletal conditions) where AI-generated content was evaluated against nuanced clinical reasoning standards that LLMs appear less equipped to meet. This pattern suggests that the most appropriate use cases for GenAI in disability services are those involving structured content generation, repetitive skill practice, or administrative efficiency—rather than those requiring clinical judgement, patient-specific contextualisation, or professional accountability.

One of the most consequential findings of this review is the concentration of evidence in professional education and the near-absence of direct patient-facing clinical evidence. While the review identified 96 studies, only a handful engaged directly with patients as users of GenAI in clinical rehabilitation contexts (Neo, Ser and Tay, 2024; Qiang *et al.*, 2025; Xiong *et al.*, 2026; Yu *et al.*, 2026), with universally small samples and no clinical outcome data beyond satisfaction or engagement. This gap is the most strategically important in the corpus: the question of whether GenAI can improve patient functional outcomes, adherence, or safety in rehabilitation remains entirely unanswered. This mirrors earlier observations that AI in rehabilitation broadly has been characterised by limited high-quality clinical evidence and an absence of longitudinal outcome data (Sumner *et al.*, 2023; Estes-Schmalzl and Lefebvre, 2025; Sadiq *et al.*, 2026).

Limitations

Several limitations of this scoping review should be acknowledged. First, the search was restricted to three databases (Scopus, Web of Science, and Lens.org) and to the Title field only. While this decision was made to ensure precision following a preliminary broader search that yielded substantial noise, it may have excluded relevant studies that address GenAI in rehabilitation or disability contexts only in their abstracts or keywords. Second, the restriction to English-language sources excluded potentially relevant research published in other languages, particularly from East Asian and Arabic-speaking contexts where GenAI research is growing rapidly. Third, grey literature was restricted to sources indexed within the three searched databases; dedicated repositories, institutional reports, and policy documents were not searched, consistent with the acknowledged challenges of achieving comprehensive grey literature coverage in scoping reviews (Khalil *et al.*, 2021).

Fourth, data extraction was conducted by a single reviewer (the principal investigator) using an AI-assisted platform (Elicit), with corrections applied where extraction was incomplete or imprecise. While screening reliability was strong ($\kappa = 0.938$), the absence of independent dual extraction for the charting process introduces a risk of extraction error or inconsistency that dual extraction would have mitigated. Fifth, consistent with JBI scoping review methodology (Aromataris *et al.*, 2024), no formal critical appraisal of included sources was conducted. This means that the synthesis presents findings from studies of varying methodological quality without weighting by rigour, and the results should not be interpreted as evidence of effectiveness. Sixth, the rapid pace of GenAI development means that the evidence base is evolving faster than traditional publication timelines allow; studies published after May 2026 are not captured, and the applicability of findings from earlier model versions to current tools may be limited.

Finally, as with all evidence syntheses in active publication areas, publication bias and positive-outcome bias cannot be excluded. Studies reporting positive or statistically significant effects of GenAI tools are likely to have been published and indexed more readily than null or negative findings, and several of the included conference papers and emerging-venue publications may not have undergone the same level of peer scrutiny as journal articles. Formal assessment of publication bias (e.g. funnel plots, Egger's test) is neither applicable nor recommended for scoping reviews, but the reader should interpret the predominantly positive tone of the included literature with this caveat in mind. The restriction to English-language sources additionally introduces a language bias that may have excluded relevant research published in Korean, Chinese, Arabic, and other languages—populations particularly relevant to this review's global scope.

Finally, while this review mapped the breadth of the evidence base, it did not examine the quality of individual study reporting or assess the risk of bias within intervention studies. Future systematic reviews targeting specific interventions, populations, or outcomes within this field should incorporate formal quality assessment and, where sufficient homogeneity exists, quantitative synthesis.

CONCLUSIONS

This scoping review provides the first comprehensive mapping of the evidence on GenAI and LLM applications in rehabilitation and disability services. Across 96 studies, the evidence consistently demonstrates that GenAI holds genuine potential for enhancing rehabilitation education, supporting special education practice, and improving accessibility for persons with disabilities—but only under specific conditions. The most robust benefits are associated with structured, educator-guided applications where GenAI supplements rather than replaces human instruction, with deliberate pedagogical design and clearly defined task parameters. Unstructured or unsupervised use does not produce equivalent outcomes and may carry risks of cognitive offloading, accuracy failures, and the reinforcement of existing inequities.

The evidence base does not yet support conclusions about patient outcomes, long-term competency development, or population-level equity effects. The near-total absence of longitudinal data, the concentration of evidence in professional education settings, and the sparse representation of direct patient-facing clinical applications represent critical gaps that must be addressed before GenAI can be recommended for broader deployment in rehabilitation service delivery. Equally important, over half of the included studies did not specify the disability type addressed, and several populations central to rehabilitation services—including persons with traumatic brain injury, spinal cord injury, progressive neurological conditions, and complex communication needs—remain virtually unrepresented.

Four priorities emerge for the field. First, longitudinal studies are needed to evaluate sustained impacts of GenAI on clinical competency, learning outcomes, and functional rehabilitation goals. Second, direct patient-facing clinical research—with clinical outcome measures beyond satisfaction—is urgently required to determine whether GenAI integration into rehabilitation workflows improves patient outcomes and safety. Third, research must extend to under-represented disability populations, younger children, older adults, and low- and middle-income country settings where rehabilitation service gaps are most acute (Kayola *et al.*, 2023). Fourth, disability-inclusive AI governance frameworks—developed with the direct participation of persons with disabilities as co-creators rather than passive subjects—are needed to ensure that GenAI deployment in rehabilitation and disability services advances equity rather than reinforcing existing barriers. Additional priorities include cost-effectiveness analyses, clinician and educator acceptance studies, patient safety monitoring, and explainable AI approaches suited to rehabilitation and disability contexts.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

AUTHOR CONTRIBUTIONS

A.A., A.A.A., and M.O.A.: conceptualisation; A.A.A. and M.O.A.: data curation; A.A.: formal analysis; A.A., A.A.A., and M.O.A.: funding acquisition; A.A., A.A.A., and M.O.A.: investigation; A.A.: methodology; A.A.A. and M.O.A.: project administration; A.A., A.A.A., and M.O.A.: resources; A.A.: software; A.A.A. and M.O.A.: supervision; A.A., A.A.A., and M.O.A.: validation; A.A.: visualisation; A.A., A.A.A., and M.O.A.: writing—original draft; A.A., A.A.A., and M.O.A.: writing—review & editing. All authors have read and agreed to the published version of the manuscript.

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DATA AVAILABILITY STATEMENT

No new data were created or analysed in this study. The supplementary files supporting the reported search strategy, data charting, and PRISMA-ScR compliance (Supplementary Files S1-S3) are openly available via the Open Science Framework (OSF): https://osf.io/s3tgz/overview?view_only=14a1cbcb558e459689ce2314b9625491.

CODE AVAILABILITY STATEMENT

The Python (version 3.12) scripts used for computational deduplication of database records are available from the corresponding author upon reasonable request. Screening and data extraction were conducted using Elicit (<https://elicit.com>) and Claude Code (Anthropic; <https://claude.ai>), which are commercial AI platforms; no bespoke model code was developed or archived.

ETHICAL APPROVAL

Ethical approval was unnecessary because the study involved neither human nor non-human participants.

DECLARATION OF INTEREST STATEMENT

Informed consent was not required because the study did not involve human or non-human participants.

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